6. ADAPTIVE FUZZY CONTROL

A convinient, alternative method to control (unknown) nonlinear systems

Pro: It is applicable with realtively easy mathematical treatment
Con: No exact guarrantee for stability (but can be ensured with high probability)

Adaptive control: In order to decrease the control error, the parameters of the controller are tuned.

Adaptive fuzzy control: Fuzzy systems is/are used in the controller whose parameters (in the input, output membership functions),. i.e. the rules are tuned in order to impove performance.

Assumed system type:
- SISO (Single input, single output)
- MIMO (Multiple input, multiple output)

The characteristic of control
- Indirect: Create model about the unknown system (identification), then a controller is design based ont he model.
- Direct: The controil system is determined directly without using any system model

Options for parameter tuning:
- 1st type: parameters are tuned only in the output membership functions
- 2nd type: Every parameter is tuned in every membership functions

6.1. SISO adaptive fuzzy control

Differential equation of the assumed system: $y^{(n)} = f(x) + g(x)u$

$u \in R^1,\ y \in R^1,\ g(x) > 0$

State vector: $x = (y, y', \dots, y^{(n-1)})^T$

Refernce signal: $y_d(t)$

Control error: $e = y_d - y$

vector of error derivatives: $\tilde{e} = (e, e', \dots, e^{(n-1)})^T$

Refernce model of the closed loop error: $e^{(n)} + k_1 e^{(n-1)} + \dots + k_n e = 0$

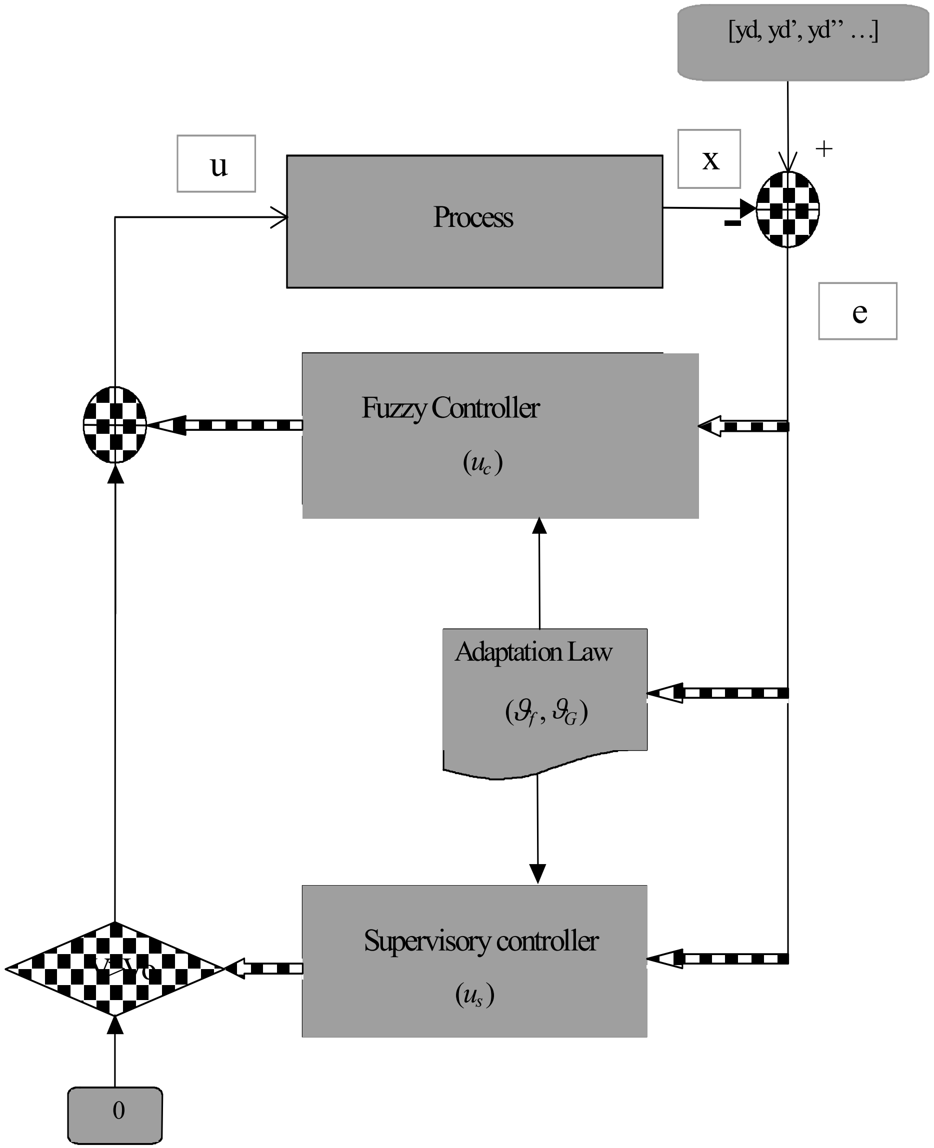
$\tilde{k} = (k_n, k_{n-1}, \dots, k_1)^T.$

where k_i is chosen such that the characteristic equation

$$s^n + k_1 s^{n-1} + \cdots + k_n = 0$$

ensures stability (Hurwitz polynomial) and fast transients.

Az önhangoló adaptív fuzzy irányítás struktúrája:



6.1.1. Control of the known system

Assumptions: The system to be controlled is known precisely.

Remark: we consider the plant (functions $f(\cdot)$ and $g(\cdot)$) known only in this subsection.

The goal in the next part of the discussion is to contruct control if the plant is not known (e.g. by approximating functions $f(\cdot)$ and $g(\cdot)$)

Let the **control law** (the nominal control signal):

$$\boxed{u = \frac{1}{g}\left(y_d^{(n)} - f + < \tilde{k}, \tilde{e} >\right)} \qquad \text{(NC)}$$

which results

$$y^{(n)} = f + g \cdot \frac{1}{g}\left(y_d^{(n)} - f + < \tilde{k}, \tilde{e} >\right) = y_d^{(n)} + < \tilde{k}, \tilde{e} >$$

to the nth derivative of the output. It yields

$$e^{(n)} = y_d^{(n)} - y^{(n)} = - < \tilde{k}, \tilde{e} > \Rightarrow$$
$$e^{(n)} + k_1 e^{(n-1)} + \ldots + k_n e = 0$$

which **ensures that control error tends to zero exponentially.**

6.1.2. Indirect adaptive fuzzy control

6.1.2.1. Concept

Functions specifying the real plant: $f(\cdot),\ g(\cdot)$ $\rightarrow$ unknown

Plant is approximated by a (fuzzy) model: $\hat{f}(\cdot),\ \hat{g}(\cdot)$

The control law:

$$\boxed{u_c = \frac{1}{\hat{g}}\left(y_d^{(n)} - \hat{f} + <\tilde{k},\tilde{e}>\right)} \qquad \text{(IAFC)}$$

Remark: If $\hat{f}(\cdot),\ \hat{g}(\cdot)$ were exactly equivalent to $f(\cdot),\ g(\cdot)$ (ideal case) $\rightarrow$ converges to nominal control (NC) and the desired exponential control error.

In case of unprecise approximation (real case), however:

$$y^{(n)} = f + g\,\hat{g}^{-1}\left(y_d^{(n)} - \hat{f} + <\tilde{k},\hat{e}>\right)$$

$$\begin{aligned} e^{(n)} &= y_d^{(n)} - y^{(n)} = y_d^{(n)} - f - g\,\hat{g}^{-1}\left(y_d^{(n)} - \hat{f} + <\tilde{k},\tilde{e}>\right) = \\ &= y_d^{(n)} - f - g\,\hat{g}^{-1}\left(y_d^{(n)} - \hat{f} + <\tilde{k},\tilde{e}>\right) + \hat{g}\,\hat{g}^{-1}\left(\hat{f} - \hat{f} + <\tilde{k},\tilde{e}> - <\tilde{k},\tilde{e}>\right) = \\ &= -<\tilde{k},\tilde{e}> + \hat{f} - f + (\hat{g} - g)\hat{g}^{-1}\left(y_d^{(n)} - \hat{f} + <\tilde{k},\tilde{e}>\right) = \\ &= -<\tilde{k},\tilde{e}> + \hat{f} - f + (\hat{g} - g)u_c, \end{aligned}$$

casting it into state space form:

$$\begin{aligned}
\dot{\tilde{e}}_1 &= \tilde{e}_2 \\
&\vdots \\
\dot{\tilde{e}}_{n-1} &= \tilde{e}_n \\
\dot{\tilde{e}}_n &= -<\tilde{k},\hat{e}> + \hat{f} - f + (\hat{g} - g)u_c.
\end{aligned}$$

which can be described in matrix form:

$$\dot{\tilde{e}} = \underbrace{\begin{bmatrix} 0 & 1 & & \\ & & \ddots & \\ & & & 1 \\ -k_n & -k_{n-1} & & -k_1 \end{bmatrix}}_{\Lambda_c} \tilde{e} + \underbrace{\begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}}_{b_c} \{\hat{f} - f + (\hat{g} - g)u_c\}$$

$$\dot{\tilde{e}} = \Lambda_c \tilde{e} + b_c \{(\hat{f} - f) + (\hat{g} - g)u_c\} \qquad \text{(HD)}$$

Remark: The resulting system is nonlinear because of $(\hat{f} - f) + (\hat{g} - g)u_c$, and sensitive to the approximation error!

Concept:

Goal:	Ensure stability
Approach:	Lyapunov stability theory
The method for stabilization:	Supervisory control activate above a predefined error level

6.1.1.2. Stability issues.

Error dynamics (HD): $\dot{\tilde{e}} = \Lambda_c \tilde{e} + b_c \{(\hat{f} - f) + (\hat{g} - g) u_c\}$

The linear part: $\dot{\tilde{e}} = \Lambda_c \tilde{e}$

Theorem [Lyapunov stability theorem for linear system]. A linear system is stable if for $Q > 0$ positive definit (and symmetrical) matrix the Lyapunov equation has a $P > 0$ positive definit (and symmetrical) solution:

$$\Lambda_c^T P + P \Lambda_c = -Q, \quad Q > 0,$$

Remark. The linear part in our case is fictitious, thus it does not provide sufficient condition for stability.One needs the general results of Lyapunov stability theorem (elaborated to nonlinear systems):

Theorem [Lyapunov stability theorem for nonlinear system]. The *nonlinear system* is asymptotically stable if for positive definite Lyapunov function V the derivative of the Lyapunov function $\frac{dV}{dt}$ is negative definite along the system trajectory (state equation solution).

Remark. If the error and its derivatives are large then $\tilde{e}$ is large, and the Lyapunov function

$$V = \frac{1}{2} < P\tilde{e}, \tilde{e} >$$

is also large. Hence if we want to keep the error bounded then we should make V bounded.

6.1.1.3. Design of supervisory control (u_s)

Goal: Keep a given V within a bound V_0.

Remark. We assume that approximation $\hat{f}(\cdot)$, $\hat{g}(\cdot)$ are sufficiently accurate within bound V_0 to consider the linear part dominant in the error dynamics which guarrantees assymptotic stability via u_c, and Λ_c.

Strategy int he supervisory control:

$$\boxed{V \text{ bounded} \Leftrightarrow \text{ if } V > V_0 \text{ then } \frac{dV}{dt} < 0}$$

Explanation: If the error increases above bound V_0, the supervisory control switches on, and enforce V to decrease bellow V_0 again.

Complete control law in case of $V > V_0$: $u = u_c + u_s$

Error dynamics (with supervisory control):

$$y^{(n)} = f + g\,u_c + g\,u_s = f + g\,\hat{g}^{-1}\left(y_d^{(n)} - \hat{f} + <\tilde{k}, \hat{e}>\right) + gu_s,$$
$$e^{(n)} = -<\tilde{k}, \hat{e}> + \hat{f} - f + (\hat{g} - g)u_c - gu_s,$$
$$\dot{\tilde{e}} = \Lambda_c \tilde{e} + b_c\{(\hat{f} - f) + (\hat{g} - g)u_c - gu_s\} \qquad \text{(HDS)}$$

The derivative of Lyapunov function (to be made negative by u_s):

$$\frac{dV}{dt} = \frac{1}{2} < P\dot{\tilde{e}}, \tilde{e} > + \frac{1}{2} < P\tilde{e}, \dot{\tilde{e}} >=$$
$$= \frac{1}{2} < P[\Lambda_c \tilde{e} + b_c\{(\hat{f} - f) + (\hat{g} - g)u_c - gu_s\}], \tilde{e} > +$$
$$+ \frac{1}{2} < P\tilde{e}, [\Lambda_c \tilde{e} + b_c\{(\hat{f} - f) + (\hat{g} - g)u_c - gu_s\}] >=$$
$$= \frac{1}{2} < (\Lambda_c^T P + P\Lambda_c)\tilde{e}, \tilde{e} > + < b_c^T P\tilde{e}, \hat{f} - f + (\hat{g} - g)u_c > - < b_c^T P\tilde{e}, gu_s >=$$
$$= -\frac{1}{2} < Q\tilde{e}, \tilde{e} > + < b_c^T P\tilde{e}, \hat{f} - f + (\hat{g} - g)u_c > - < b_c^T P\tilde{e}, gu_s >$$

(LFD)

[Back to ujLFD]

Assumption: Bounds g_L, g_U, f_U are known for which hold

- $0 < g_L \le g(x) \le g_U$,
- $|f(x)| \le f_U$.

We are looking for supervisory control u_s that ensures negative derivative of Lyapunov function. Contemplating about [(LFD)], one may choose

$$\boxed{u_s := g_L^{-1} b_c^T P \tilde{e} \left\{ |\hat{f}| + f_U + |\hat{g}u_c| + |g_U u_c| \right\} / |b_c^T P\tilde{e}|} \qquad \text{(SC)}$$

The derivative of Lyapunov function with the chosen supervisory control:

$$\frac{dV}{dt} \le -\frac{1}{2} < Q\tilde{e}, \tilde{e} > + \left| b_c^T P \tilde{e} \right| \left\{ \left| \hat{f} \right| + \left| f \right| + \left| \hat{g} u_c \right| + \left| g u_c \right| \right\} - < b_c^T P \tilde{e}, g\, g_L^{-1}\, b_c^T P \tilde{e} \left\{ \left| \hat{f} \right| + f_U + \left| \hat{g} u_c \right| + \left| g_U u_c \right| \right\} / \left| b_c^T P \tilde{e} \right. > \Big| \qquad \text{(LFDS)}$$

in SISO case $g\, g_L^{-1} \ge 1$ and $b_c^T P \tilde{e}$ is scalar valued, hence the 3rd term of dV / dt can be rewritten:

$$-\left| b_c^T P \tilde{e} \right|^2 \cdot g\, g_L^{-1} \left\{ \left| \hat{f} \right| + f_U + \left| \hat{g} u_c \right| + \left| g_U u_c \right| \right\} / \left| b_c^T P \tilde{e} \right| =$$
$$-\left| b_c^T P \tilde{e} \right| \cdot g\, g_L^{-1} \left\{ \left| \hat{f} \right| + f_U + \left| \hat{g} u_c \right| + \left| g_U u_c \right| \right\}.$$

The sum of the last two terms in (LFDS) are negative since

$$\left| b_c^T P \tilde{e} \right| \left\{ \left| \hat{f} \right| + \left| f \right| + \left| \hat{g} u_c \right| + \left| g u_c \right| \right\} - \left| b_c^T P \tilde{e} \right| \cdot g\, g_L^{-1} \left\{ \left| \hat{f} \right| + f_U + \left| \hat{g} u_c \right| + \left| g_U u_c \right| \right\} \le$$
$$\le \left| b_c^T P \tilde{e} \right| \left\{ \left| f \right| - f_U + (g - g_U) \left| u_c \right| \right\} \le 0,$$

which means that

$$\frac{dV}{dt} \le -\frac{1}{2} < Q\tilde{e}, \tilde{e} >$$

In other words, supervisory control ensures $\dfrac{dV}{dt} < 0$.

Remark. The derivation above can be applied to any approximation of $\hat{f}(\cdot)$, $\hat{g}(\cdot)$ (not only to fuzzy approx.)

6.1.1.4. 1st type indirect adaptive fuzzy control

Assumption. In the model of the dynamic system, the functions $f(\cdot)$, $g(\cdot)$ are described by zero order Sugeno systems:

$$\hat{f} = \sum_i y^i \frac{w_i}{\sum_k w_k} = \sum_i y^i \varphi_i(x)$$

The parameters of the antecedent parts are fixed, and the only tunable parameters are y^i, $i = 1, \ldots, m$ in the consequent parts.

Notations:

$$\vartheta_f = (y^1, \ldots, y^m)^T$$
$$\varphi^T = (\varphi_1, \ldots \varphi_m)$$

The approximation of the functions: $\hat{f}(x) = \varphi^T(x)\vartheta_f$

$$\hat{g}(x) = \psi^T(x)\vartheta_g$$

(Is it familiar with the linear parameter estimation???)

Based on Wang's theorem, any continuous function over a compact set can be well approximated by zero order Sugeno systems for any prescribed precision. Let them be:

$$\vartheta_f^* = \arg\min_{\vartheta_f} \sup_x \left| \hat{f}(x) - f(x) \right| \Rightarrow \hat{f}^*(x),$$
$$\vartheta_g^* = \arg\min_{\vartheta_g} \sup_x \left| \hat{g}(x) - g(x) \right| \Rightarrow \hat{g}^*(x),$$

Introducing: $$w(x) := \hat{f}(x, \vartheta_f^*) - f(x) + [\hat{g}(x, \vartheta_g^*) - g(x)]u_c$$

Remark. Based on Wang's theorem $w(\cdot)$ can be made theoretically small if the complexity of the fuzzy systems (the number of relations) is large enough (It plays important role int he proof of stability).

Let $\Delta\vartheta_f := \vartheta_f - \vartheta_f^*$ $\qquad$ $\Delta\vartheta_g := \vartheta_g - \vartheta_g^*$

Then the previously derived error dynamics reads

$$\dot{\tilde{e}} = \Lambda_c \tilde{e} + b_c\{(\hat{f} - f) + (\hat{g} - g)u_c\} \qquad \text{(HD)}$$

or

$$\dot{\tilde{e}} = \Lambda_c \tilde{e} + b_c\{(\hat{f} - f) + (\hat{g} - g)u_c - gu_s\} \qquad \text{(HDS)}$$

The terms in error dynamics can be rewritten

$$\begin{aligned} &\hat{f} - f + (\hat{g} - g)u_c = \hat{f} - \hat{f}^* + (\hat{g} - \hat{g}^*)u_c + w \\ &= \varphi^T(x)\Delta\vartheta_f + \psi^T(x)\Delta\vartheta_g u_c + w = \varphi^T(x)\Delta\vartheta_f + \psi^T(x)u_c\Delta\vartheta_g + w. \end{aligned} \qquad (1)$$

Hence the new form of **error dynamics (HDS):**

$$\dot{\tilde{e}} = \Lambda_c \tilde{e} + b_c[\varphi^T(x)\Delta\vartheta_f + \psi^T(x)u_c\Delta\vartheta_g + w] - gu_s$$

<u>Derivation of adaptation/tuning law</u>

Concept: Seek a Lyapunov function which penalizes the deviation from optimal parameters (we want to have stable controller)

$$V := \frac{1}{2} < P\tilde{e}, \tilde{e} > + \frac{1}{2\gamma_1} < \Delta\vartheta_f, \Delta\vartheta_f > + \frac{1}{2\gamma_2} < \Delta\vartheta_g, \Delta\vartheta_g > \qquad (2)$$

where $\gamma_1 > 0$, $\gamma_2 > 0$ appropriately chosen weighting constants (learning speed).

Exploitimng the derivation of the first term (LFD) in (2)

$$\frac{dV}{dt} = -\frac{1}{2} < Q\tilde{e}, \tilde{e} > + < b_c^T P\tilde{e}, \varphi^T(x)\Delta\vartheta_f + \psi^T(x)u_c\Delta\vartheta_g > + < b_c^T P\tilde{e}, w > -$$

$$- < b_c^T P\tilde{e}, u_s > + \frac{1}{\gamma_1} < \Delta\dot{\vartheta}_f, \Delta\vartheta_f > + \frac{1}{\gamma_2} < \Delta\dot{\vartheta}_g, \Delta\vartheta_g > =$$

$$= -\frac{1}{2} < Q\tilde{e}, \tilde{e} > - < b_c^T P\tilde{e}, u_s > + < b_c^T P\tilde{e}, w > +$$

$$+ < \varphi(x) b_c^T P\tilde{e} + \frac{1}{\gamma_1}\Delta\dot{\vartheta}_f, \Delta\vartheta_f > + < \psi(x)u_c b_c^T P\tilde{e} + \frac{1}{\gamma_2}\Delta\dot{\vartheta}_g, \Delta\vartheta_g >$$

The last two terms can be made zero by choosing suitable $\Delta\dot{\vartheta}_f$ and $\Delta\dot{\vartheta}_g$ hence one can rid off from two terms in dV/dt that might influence the value in positive direction. Since $\Delta\vartheta_f := \vartheta_f - \vartheta_f^*$ and ϑ_f^* do not depend on time, $\Delta\dot{\vartheta}_f = \dot{\vartheta}_f$. Similarly, $\Delta\dot{\vartheta}_g = \dot{\vartheta}_g$. It means that one can use the adaptation law:

$$\boxed{\begin{aligned} \dot{\vartheta}_f &= -\gamma_1 \varphi(x) b_c^T P\tilde{e} \\ \dot{\vartheta}_g &= -\gamma_2 \psi(x) u_c b_c^T P\tilde{e} \end{aligned}}$$

The derivative of the Lyapunov function with the adaptation law:

$$\frac{dV}{dt} = -\frac{1}{2} < Q\tilde{e}, \tilde{e} > - < b_c^T P\tilde{e}, u_s > + < b_c^T P\tilde{e}, w >$$

Remarks. The smaller the approximation error $w(\cdot)$, the higher the possibility that dV/dt is negative and the control system is stable.

Remarks:

1. Supervisory control ensures only to lead back the error into an error band (with zero mean) buti t does not ensure stability!
2. Stability needs a negative deravative of Lyapunov function within bound V_0 (Supervisory control switched off at this time) using a given number of rules while the value of $w(\cdot)$ is unknown. (If supervisory control switched on than the derivative of Lyapunov function is negative by design concept in case of known limits g_L, g_U, f_U)
3. In case of small V_0, the control shows the characteristic of sliding mode control. Then bang-bang effect may occur (chattering, oscillation), since $\hat{f}(\cdot)$, $\hat{g}(\cdot)$ never tuned. However it was assumed that the approximation is good dominates error dynamics within the Lyapunov function bound V_0.

Luenberger projection

Goal: Keep the parameters inside a sphere $|\vartheta| \le R$ (where ϑ may be ϑ_f or ϑ_g)

Concept: If the new parameter leaves the sphere during parameter change, the gradient can be projected onto the tangential plane (Luenberger projection)

Let $-g$ be the negative gradient (direction of parameter change)

Since the normal vector of the tangential plane in point ϑ has the direction of the radius of the sphere to this point, hence the unite normal vector is $\vartheta / |\vartheta|$ and the projection of $-g$ onto the normal vector is:

$$< -g, \frac{\vartheta}{|\vartheta|} > \frac{\vartheta}{|\vartheta|}$$

The corrected $-g_c$ is the projection of $-g$ onto the tangential plane:

$$\boxed{-g_c := -g - < -g, \frac{\vartheta}{|\vartheta|} > \frac{\vartheta}{|\vartheta|} = -g + \vartheta \frac{< \vartheta, g >}{|\vartheta|^2} = -g + \frac{\vartheta \vartheta^T}{|\vartheta|^2} g}$$

Hence the adaptation law will be modified if the new parameter would leave the constraint sphere:

$$\boxed{\begin{aligned} \dot{\vartheta}_f &= -\gamma_1 \varphi(x) b_c^T P\tilde{e} + \frac{\vartheta_f \vartheta_f^T}{|\vartheta_f|^2} \gamma_1 \varphi(x) b_c^T P\tilde{e} \\ \dot{\vartheta}_g &= -\gamma_2 \psi(x) u_c b_c^T P\tilde{e} + \frac{\vartheta_g \vartheta_g^T}{|\vartheta_g|^2} \gamma_2 \psi(x) u_c b_c^T P\tilde{e} \end{aligned}}$$

6.1.1.5. 2nd type indirect fuzzy control

Concept: The parameters of the membership functions in the antecedent parts will also be tuned.

Principle: Since $\hat{f}(\cdot)$ and $\hat{g}(\cdot)$ depend nonlinearly of these parameters, we take the Taylor series of them and leave the higher order terms. Let consider for example $\hat{f}(\cdot)$, then:

$$\hat{f}^* := \hat{f}(x,\vartheta_f^*) \approx \hat{f}(x,\vartheta_f) + < \frac{d\hat{f}}{d\vartheta_f}, \vartheta_f^* - \vartheta_f > \qquad \text{(Taylor series around } \vartheta_f\text{)}$$

Similar results can be derived to $\hat{g}^*(\cdot)$. Then expression (1) can be rewritten as

$$\hat{f} - f + (\hat{g} - g)u_c = \hat{f} - \hat{f}^* + (\hat{g} - \hat{g}^*)u_c + w \approx < \frac{d\hat{f}}{d\vartheta_f}, \Delta\vartheta_f > + < \frac{d\hat{g}}{d\vartheta_g}, \Delta\vartheta_g > + w$$

where $\varphi^T(x)$ and $\psi^T(x)$ in (1) corresponds to

$$\varphi^T(x) := \left(\left. \frac{d\hat{f}}{d\vartheta_f} \right|_{\vartheta_f} \right)^T \qquad \text{and} \qquad \psi^T(x) := \left(\left. \frac{d\hat{g}}{d\vartheta_g} \right|_{\vartheta_g} \right)^T \qquad \text{(defined as row vectors)}$$

hence the adaptation law exploiting the results of 1st type controller:

$$\boxed{\begin{aligned} \dot{\vartheta}_f &= -\gamma_1 \frac{d\hat{f}}{d\vartheta_f} b_c^T P\tilde{e} \\ \dot{\vartheta}_g &= -\gamma_2 \frac{d\hat{g}}{d\vartheta_g} u_c b_c^T P\tilde{e} \end{aligned}}$$

6.1.3. Direct adaptive fuzzy control

The nominal control for known plant: $u^* = g^{-1}[y_d^{(n)} - f + <\tilde{k}, \tilde{e}>]$ (IKDF)

[link:bound]

However $f(\cdot)$ and $g(\cdot)$ are not known, hence this control would be the ideal case.

Similarly to indirect case, the complete control consits of two components:

$$u = u_c + u_s$$

u_c - Fuzzy direct control

u_s - Supervisory control

6.1.3.1. Supervisory control for direct adaptive (fuzzy) control

The control signal is described in the following grouping:

$$u = u_c + u_s + u^* - u^* = u^* + (u_c + u_s - u^*)$$

we follow the same path to design a supervisory control for stabiulity, but Lyapunov function does NOT contain approximations $\hat{f}(\cdot)$ and $\hat{g}(\cdot)$. Substituting the control into the dynamic equation

$$y^{(n)} = y_d^{(n)} + < \tilde{k}, \tilde{e} > + g(u_c + u_s - u^*)$$
$$e^{(n)} = y_d^{(n)} - y^{(n)} = - < \tilde{k}, \tilde{e} > + g(u^* - u_c - u_s)$$
$$\dot{\tilde{e}} = \Lambda_c \tilde{e} + b_c g(u^* - u_c - u_s),$$

Taking again the Lyapunov equation:

$$\Lambda_c^T P + P\Lambda_c = -Q, \quad Q > 0,\ P > 0$$ (P is a positive definite symmetric solution)

Using the solution, one can construct the Lyapunov function $V = \frac{1}{2} < P\tilde{e}, \tilde{e} >$. Its derivative is

$$\frac{dV}{dt} = \frac{1}{2} < P\dot{\tilde{e}}, \tilde{e} > + \frac{1}{2} < P\tilde{e}, \dot{\tilde{e}} > = \frac{1}{2} < P[\Lambda_c \tilde{e} + b_c g(u^* - u_c - u_s)], \tilde{e} > + \frac{1}{2} < P\tilde{e}, \Lambda_c \tilde{e} + b_c g(u^* - u_c - u_s) > =$$
$$= \frac{1}{2} < (\Lambda_c^T P + P\Lambda_c)\tilde{e}, \tilde{e} > + < b_c^T P\tilde{e}, g(u^* - u_c) > - < b_c^T P\tilde{e}, g u_s > .$$

Similarly to previous case, bounds $0 < g_L \le g(x) \le g_U$ and $|f(x)| \le f_U$ are introduced, hence considering (IKDF)

$$|u^*| \le g_L^{-1}[|y_d^{(n)}| + f_U + |<\tilde{k},\tilde{e}>|],$$

$$g|u^*| \le gg_L^{-1}[|y_d^{(n)}| + f_U + |<\tilde{k},\tilde{e}>|],$$

$$\frac{dV}{dt} \le -\frac{1}{2} < Q\tilde{e},\tilde{e} > + |b_c^T P\tilde{e}| \left\{ g|u_c| + g|u^*| \right\} - < b_c^T P\tilde{e}, gu_s >$$

$$\frac{dV}{dt} \le -\frac{1}{2} < Q\tilde{e},\tilde{e} > + |b_c^T P\tilde{e}|\, g \left\{ |u_c| + g_L^{-1}[|y_d^{(n)}| + f_U + |<\tilde{k},\tilde{e}>|] \right\} - < b_c^T P\tilde{e}, gu_s >$$

It is known that $gg_L^{-1} \ge 1$, hence using

$$\boxed{u_s := sign(b_c^T P\tilde{e}) \left\{ |u_c| + g_L^{-1} \left[|y_d^{(n)}| + f_U + |<\tilde{k}\tilde{e},\tilde{e}>| \right] \right\} \quad \text{if } V > V_0}$$

if $V > V_0$ the derivative of Lyapunov function

$$\frac{dV}{dt} \le -\frac{1}{2} < Q\tilde{e},\tilde{e} > + |b_c^T P\tilde{e}|\, g \left\{ |u_c| + g_L^{-1}[|y_d^{(n)}| + f_U + |<\tilde{k},\tilde{e}>|] \right\} -$$
$$- |b_c^T P\tilde{e}| g \left\{ |u_c| + g_L^{-1}[|y_d^{(n)}| + f_U + |<\tilde{k},\tilde{e}>|] \right\} \le$$
$$\le -\frac{1}{2} < Q\tilde{e},\tilde{e} >.$$

is negative, which keeps the error within a specified bound.

6.1.3.2. 1st type direct fuzzy control

Nominal control: Zero order Sugeno system

$u_c = \varphi^T(x)\vartheta$ (ϑ contains only the parameters in consequent part of the relations $\vartheta_k = \bar{y}^k$)

Linear in parameters

$$\varphi_k^T(x\,|\,\theta) = \frac{\prod_{i=1}^{n} \exp\left(-\left(\frac{x_i - \bar{x}_i^k}{\sigma_i^k}\right)^2\right)}{\sum_{l=1}^{M}\prod_{i=1}^{n} \exp\left(-\left(\frac{x_i - \bar{x}_i^l}{\sigma_i^l}\right)^2\right)}$$

The approximation of ideal $u^*(x)$ with the best zero order Sugeno system is:

$$\vartheta^* := \arg\min_{\vartheta} \sup_{x} \left|\varphi^T(x)\vartheta - u^*(x)\right| \Rightarrow \boxed{u_c^*(x) = \varphi^T(x)\vartheta^*}$$

Then

$$w := u*(x) - u_c^*(x)$$

$$u^* - u_c = u^* - u_c^* + u_c^* - u_c = w - \varphi^T(x)(\vartheta^* - \vartheta) = w - \varphi^T(x)\Delta\vartheta$$

$$\dot{\tilde{e}} = \Lambda_c\tilde{e} + b_c g\{w - \varphi^T\Delta\vartheta - u_s\}$$

The derivation of the adaptation law.

Let the Lyapunov function be: $V = \frac{1}{2} < P\tilde{e}, \tilde{e} > + \frac{1}{2\gamma} < \Delta\vartheta, \Delta\vartheta >$ (the parameter error is penalized)

Its derivative:

$$\frac{dV}{dt} = \frac{1}{2} < P\dot{\tilde{e}}, \tilde{e} > + \frac{1}{2} < P\tilde{e}, \dot{\tilde{e}} > + \frac{1}{\gamma} < \Delta\dot{\vartheta}, \Delta\vartheta > =$$

$$= \frac{1}{2} < P[\Lambda_c \tilde{e} + b_c g\, w - b_c g\, \varphi^T \Delta\vartheta - b_c g\, u_s], \tilde{e} > + \frac{1}{2} < P\tilde{e}, \Lambda_c \tilde{e} + b_c g\, w - b_c g\, \varphi^T \Delta\vartheta - b_c g\, u_s > + \frac{1}{\gamma} < \Delta\dot{\vartheta}, \Delta\vartheta > =$$

$$= \frac{1}{2} < (\Lambda_c^T P + P\Lambda_c)\tilde{e}, \tilde{e} > + \underbrace{< b_c^T P\tilde{e}, gw >}_{Ordo(w)} \underbrace{- \underbrace{< b_c^T P\tilde{e}, gu_s >}_{\text{pozitív}}}_{\text{negatív}} + \frac{1}{\gamma} < \Delta\dot{\vartheta} - \gamma\, \varphi\, g\, b_c^T P\, \tilde{e}, \Delta\vartheta > .$$

The last term can be vanished by appropriate tuning of the parameter vector:

$$\Delta\dot{\vartheta} = \dot{\vartheta} := \gamma\, \varphi\, g\, b_c^T P\, \tilde{e}$$

However $g(x)$ is unknown, therefore the problem is simplified:

Assumption: $0 < g_L \le g(x) = konst. = g_0$ (g_0 is unknown)

Adaptation/tuning law:

$$\boxed{\dot{\vartheta} := \underbrace{\gamma\, g_0}_{\tilde{\gamma}}\, \varphi\, b_c^T P\, \tilde{e} = \tilde{\gamma} \varphi\, b_c^T P\, \tilde{e}}$$

($\tilde{\gamma}$ is an appropriately chosen step size – positive number)

6.1.3.3. 2nd type adaptive fuzzy control

The derivation is analogous to the case of indirect control. One can also use the local linearization.

Adaptation/tuning law:

$$\boxed{\dot{\vartheta} := \tilde{\gamma} \frac{du_c}{d\vartheta} b_c^T P \tilde{e}}$$

where $u_c(x \mid \theta) = \dfrac{\sum_{l=1}^{M} \bar{y}^l \prod_{i=1}^{n} \exp\left(-\left(\dfrac{x_i - \bar{x}_i^l}{\sigma_i^l}\right)^2\right)}{\sum_{l=1}^{M} \prod_{i=1}^{n} \exp\left(-\left(\dfrac{x_i - \bar{x}_i^l}{\sigma_i^l}\right)^2\right)}$

Remarks.

- Luenberger projection is applicable
- The methods discussed here can be easily extended to first order Sugeno systems